

Evolutionary Orchestration of Agentic Fleets: Population Search, Recombination, and the Explore–Exploit Anneal

Hanzo AI — ML Systems

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Abstract

An agentic fleet is a population of candidate behaviours acting on a shared solution space. We argue that the natural control law for such a fleet is not command but *evolution*: selection, recombination, and mutation applied to the population, generation over generation. The claim is made precise by placing three folk mechanisms on one object — a probability distribution p_t over solution space — and showing they are the same dynamics at three timescales. A mean-field game (companion paper, *Mean-Field Alignment*) is the fast inner loop: each agent best-responds to the population mean and the fleet equilibrates. Ferromagnetic criticality is the thermostat: it sets how aligned the population is. A genetic algorithm (GA) is the slow outer loop: it moves p_t itself uphill on fitness. We show a GA is a fitness-driven Fokker–Planck flow — selection is replicator drift, mutation is diffusion, crossover is a mixing operator — and that when fitness is frequency-dependent, replicator drift *is* the mean-field coupling of the game. The one operator the other two lack is crossover, and it is the source of the speedup: by the schema theorem it recombines above-average building blocks, so on a *separable* landscape it assembles the global optimum in time polynomial in the number of blocks where mutation-only takes exponential time (Proposition 2); the honest boundary is deception, where crossover is disruptive. Premature convergence is exactly the ferromagnetic ordered phase (groupthink), and the cure is annealing, not quenching. We then ground the framework twice against real Hanzo systems: the GRPO rollout *group* of [1] is a GA population and its group-relative advantage is selection; AnyMoE experts are building blocks and expert recombination is crossover; the island model of [2] is a coarse-grained GA. In the agentic harness the population is a set of solution *variants*, fitness is the adversarial critic together with a green build, crossover is a synthesis agent that grafts the fittest module from each parent, and generations are the loop-until-dry rounds. We keep the accounting honest: this is a control lens, not a literal energy function, but the building-block argument is a genuine reason recombination accelerates the decomposable problems — files, modules, tools — that engineering fleets actually solve.

1 The pain: a fleet that averages when it should assemble

Run N agents at a hard problem and the two obvious control laws both disappoint. Make them independent and they duplicate each other’s dead ends; nobody’s partial success helps anybody else, and the fleet is a set of lonely random walks. Make them agree — align them to a running consensus — and they converge, but only to the *mean* of what they already believed; consensus averages, it does not invent. Neither law does the thing a research group does when it is working well: take the best idea from one line of attack, the best idea from another, and *combine* them into something better than either.

That combining operation has a name. It is crossover, and it is the defining move of a genetic algorithm. This paper’s thesis is that an agentic fleet should be run as a population under evolutionary search, that crossover is precisely the operator the “independent” and “aligned” regimes lack, and that it is fast for a concrete, checkable reason: the problems a software fleet solves are *decomposable* into modules, and crossover is the operator that exploits decomposition. We make each of those claims formal, then show the same three knobs already live inside Hanzo’s model-training loop and its agent harness — so “continually improving fleets” is not an aspiration bolted on the side but the loop the substrate already runs, named.

2 One population, three timescales

Fix a solution space $\mathcal{S}$ (policies, programs, tool implementations — whatever a unit of the fleet emits) and let the state of the fleet be a distribution $p_t \in \Delta(\mathcal{S})$. Everything below is a map $p_t \mapsto p_{t+1}$. Three such maps act at separated timescales, and separation of timescales is what lets them compose without fighting.

Definition 1 (The three loops). Inner (fast) — the game. *Holding p_t fixed, each agent chooses a best response to the induced mean field; iterating to the fixed point is a mean-field-game (MFG) Nash equilibrium. This is the subject of the companion paper [4] and is treated here as a black box that, given a population, returns an equilibrated one.*

Middle (medium) — the thermostat. *A scalar order parameter $m \in [0, 1]$ measures how concentrated p_t is (how aligned the fleet is). A coupling J (how strongly each agent heeds the mean) and a temperature T (how much independent noise each agent injects) set m through a ferromagnetic self-consistency; the companion paper shows the useful operating point is critical, $T \lesssim T_c$.*

Outer (slow) — the search. *A selection–recombination–mutation operator moves p_t toward higher fitness across generations. This is the genetic algorithm and is the subject of this paper.*

The rest of the paper studies the outer loop and its coupling to the other two. The organising fact is that all three are transport of the *same* p_t ; they differ only in the vector field they impose on it.

3 A genetic algorithm is a fitness-driven Fokker–Planck flow

Let $f : \mathcal{S} \rightarrow \mathbb{R}$ be fitness and write $\phi(p) = \mathbb{E}_{x \sim p}[f(x)]$ for the population-mean fitness. A generational GA applies, in order: (i) *selection*, reweighting p toward high f ; (ii) *crossover*, recombining sampled pairs; (iii) *mutation*, perturbing offspring. Take the weak-selection, small-step continuum limit of one generation. Selection and mutation become, respectively, a drift and a diffusion of p_t — a Fokker–Planck equation — and selection’s drift is exactly the replicator field.

Proposition 1 (Selection is replicator drift; mutation is diffusion). *In the weak-selection continuum limit, fitness-proportional (or Boltzmann, or tournament) selection on a population $x = (x_i)$ of type frequencies induces the replicator dynamics [9, 10]*

$$\dot{x}_i = x_i(f_i(x) - \phi(x)), \quad \phi(x) = \sum_j x_j f_j(x), \quad (1)$$

and mutation at rate μ adds a diffusion term $\mu \Delta x_i$ (in genotype space), giving a replicator–mutator (Fokker–Planck) flow. For frequency-independent f , mean fitness is non-decreasing along (1): $\dot{\phi} = \text{Var}_x(f) \geq 0$.

Proof sketch. Fitness-proportional selection maps $x_i \mapsto x_i f_i / \phi$; subtracting x_i and taking the small-step limit gives $\dot{x}_i = x_i(f_i - \phi) / \phi$, which is (1) after the standard time reparametrisation $dt \mapsto \phi dt$. Boltzmann and tournament selection agree to first order in selection strength (they differ only in the effective inverse temperature multiplying f). For the variance identity, $\dot{\phi} = \sum_i \dot{x}_i f_i = \sum_i x_i f_i (f_i - \phi) = \mathbb{E}[f^2] - \mathbb{E}[f]^2$ when f does not depend on x ; frequency dependence adds the cross term $\sum_i x_i f'_i \cdot \dot{x}$, which is where the game enters. $\square$

The cross term is not a nuisance; it is the bridge. When fitness is *frequency-dependent* — $f_i = f_i(x)$, an agent’s payoff depends on what the rest of the fleet is doing — (1) is evolutionary game theory, and its rest points are the Nash equilibria of the underlying game. That is the inner loop of Definition 1: **replicator selection with frequency-dependent fitness is the mean-field coupling**. The GA outer loop and the MFG inner loop are the same equation read at two speeds — selection slow, best-response fast — so composing them is legitimate, not a metaphor. Mutation supplies the diffusion that keeps p_t full-support so selection has variance to act on (Proposition 1 stalls at $\text{Var} = 0$); crossover is the operator neither (1) nor the mean field contains, and it is where the speed comes from.

4 Crossover is the speedup: compositionality and the schema theorem

A *schema* H is a subset of $\mathcal{S}$ fixing some coordinates and leaving others free; its *order* $o(H)$ is the number fixed and its *defining length* $\delta(H)$ the span between the outermost fixed coordinates. Holland’s schema theorem [7] bounds the next-generation count $m(H, t)$ of individuals matching H under proportional selection, one-point crossover rate p_c , and mutation rate p_m :

$$\mathbb{E}[m(H, t+1)] \geq m(H, t) \frac{f(H)}{\phi} \left(1 - p_c \frac{\delta(H)}{\ell - 1} - o(H) p_m \right), \quad (2)$$

where ℓ is the genome length and $f(H)$ the mean fitness of matching individuals. Read (2) as a statement about *building blocks*: a schema that is short (δ small), low-order (o small), and above average ($f(H) > \phi$) is disrupted little by crossover and grows roughly geometrically. Crossover’s job is to take two such blocks — carried by different parents — and place them in one child.

Averaging (the mean field) cannot do this: it returns a convex combination of parents, which for a combinatorial optimum is typically *worse* than either. Independent search cannot do it either: it has no channel to move a good block from one lineage to another. Crossover is the unique operator among the three that transports building blocks *between* lineages, and on a decomposable landscape that is decisive.

Proposition 2 (Separable speedup). *Let the fitness decompose additively over M disjoint blocks, $f(x) = \sum_{b=1}^M g_b(x_b)$, each block of size k with a unique optimal setting. Suppose selection preserves any block already at its optimum in some individual (each block-optimum is an above-average, short, low-order schema). Then uniform crossover assembles the global optimum in $O(M \ln M)$ expected recombination events (coupon-collector over blocks), i.e. time polynomial in M . A mutation-only search with no inter-block fitness gradient (the “royal road” class) requires expected time exponential in M to first reach the joint optimum.*

Proof sketch. Once each block’s optimal setting is present in *some* member of the population (which selection maintains, since flipping a block off its optimum is selected against), a recombination event copies a parent’s optimal block into a child with constant probability per block. Collecting all M

optimal blocks into one individual is the coupon-collector problem with M coupons: expected $MH_M = O(M \ln M)$ draws, each a recombination. For mutation-only on a flat-between-blocks (royal-road) landscape, no partial credit is visible: the search is a random walk on $\{0, 1\}^{Mk}$ whose only fitness signal is the all-optimal point, so the hitting time is $\Theta(2^{\Theta(Mk)})$ up to the fraction of the space selection can prune, which the royal-road construction makes vanishing. Hence polynomial versus exponential. $\square$

Remark 1 (The honest boundary: deception). *Proposition 2 assumes block optima are above average in isolation. A deceptive fitness [8] is built so that the block schemata leading toward the local optimum out-compete those leading toward the global one; there crossover assembles the wrong blocks faster, and the GA converges confidently to the trap. The correct fallback is mutation-and-diversity (raise T , Section 5), not more crossover. The reason the framework nonetheless earns its keep for us is empirical: the fleet’s problems — “add these three actions to that tool,” “wire this module into the registry,” “port these nine tools” — are separable by construction (one file, one module, one tool per unit of work), which is the regime where Proposition 2, not Remark 1, governs.*

5 The order parameter: premature convergence is the ordered phase

A GA’s classic failure is *premature convergence*: the population collapses onto one genotype, variance vanishes, and by Proposition 1 selection stalls — the search freezes in a local optimum. This is not a separate pathology from the companion paper’s groupthink; it is the same event. Identify the GA’s population homogeneity with the ferromagnetic order parameter m : $m \rightarrow 1$ is a fully converged population and a fully aligned fleet, one and the same ordered phase. Selection pressure is inverse temperature; strong selection is cold and quenches, weak selection is hot and drifts. The cure is therefore the cure for a quench: *anneal*. Run hot early — high mutation, weak coupling, diverse variants, so crossover has distinct building blocks to find — and cool toward criticality as fitness climbs, never all the way to $m = 1$. Simulated annealing [11] is exactly this temperature schedule, and the companion paper’s critical operating point $T \lesssim T_c$ is where the annealed GA should spend its late generations: correlated enough to converge, hot enough that a single high-fitness variant (or a single refuting critic) can still tip the population out of a bad basin. The external field — the directive — breaks the symmetry so the descent aims at the intended optimum rather than a merely-agreed one. **The GA is the search, the ferromagnetic thermostat is how you keep the search from freezing, and they share one knob: m .**

6 Embedding in enso training: the rollout group is a population

The framework is not new machinery for the training stack; it is a name for a loop the stack already runs. Group Relative Policy Optimization (GRPO) [5], implemented in the trainer of [1] (`ml/hanzo-training/src/grpo/`), samples, for each prompt, a *group* of G rollouts, scores them, and forms the group-relative advantage $A_i = (r_i - \langle r \rangle) / \text{std}(r)$ before a length-normalised policy-gradient step with an optional $k3$ KL penalty to a frozen reference. Read through Sections 3–5:

- **Population = the rollout group.** The G samples are the generation; the policy is the distribution p_t they are drawn from.

- **Selection = the group-relative advantage.** A_i is $f_i - \langle f \rangle$ with a variance normaliser — the replicator drift of (1) verbatim: rollouts above the group mean are reinforced, below are suppressed, and the group mean is the mean-field baseline. GRPO is replicator selection with a frequency-defined baseline.
- **Temperature = sampling temperature; coupling = the KL coefficient.** The $k3$ KL to the reference is the coupling that keeps the group from quenching away from a sane prior; the sampling temperature is T . Group size G sets how well $\langle r \rangle$ estimates the mean field.
- **Crossover = expert recombination.** The AnyMoE in-engine gate trainer of [1] (`amoe.rs`) trains gates over *frozen quantized experts*: the experts are reusable building blocks — and, as the companion *Streaming Mixture-of-Experts* [3] shows, independently addressable, byte-exact banks, exactly the substrate a module-level crossover needs — and the gate learns which to compose per token. Recombining experts across a population of gates is crossover on the MoE genome; the island model of [2] — independent sub-populations with occasional migration — is the coarse-grained GA that keeps that recombination from collapsing to one basin.
- **Generations = expert iteration.** Distilling the group’s winning rollouts back into **enso** (expert iteration / STaR [6]) is the generational update — the fittest phenotypes of generation t become the genotype prior of generation $t+1$; the harness of Section 7 supplies the fitness that scores them.

The prescription that follows. Because premature convergence is the ordered phase, the training knobs (G , T , KL) should be *annealed*, not fixed: start with large G , high T , low KL (a hot, diverse group that finds building blocks), and cool T and raise KL as reward climbs, holding the group near criticality rather than driving the advantage to a spike. This is the training-time reading of Section 5, and it is a config change over the existing GRPO loop, not new kernels.

7 Embedding in the agentic harness: variants, critics, and grafts

Inference-time, the fleet upgrades from *align-and-best-respond* to *align-and-best-respond-and-evolve*. The mapping is exact and already partially in place in the workflow engine that runs Hanzo’s swarms:

GA operator	Harness realisation
population	N solution <i>variants</i> , not merely N agents on one attempt
fitness f	adversarial critics + a green build + the shipped-live outcome
selection	keep the variants the critics do not refute
crossover	a <i>synthesis</i> agent: split candidates into modules, graft the fittest of each
mutation	a fresh-lens agent perturbing one module to escape a local optimum
generations	the loop-until-dry rounds; the completeness gap is the fitness gap
temperature	anneal coupling/selection each round: diverse early, aligned late

Crucially this fitness is largely *ground truth*, not a learned reward model that can be gamed: the build either compiles or it does not, the tests either pass or they do not, and the ultimate signal is whether the change ships and survives in production. That is what makes the harness landscape verifiable and its separable component real, the low-epistasis regime Proposition 2 needs.

Note the economy: *the same critic is three things at once* — the ferromagnetic fluctuation that stops groupthink (Section 5), the GA’s selection operator (Section 3), and, when its verdict

is a scalar reward, the training signal of Section 6. One mechanism, three roles, which is why the flywheel closes.

A worked instance. A hard tool port — a browser tool with a ~ 90 -action surface and a persistent-session requirement — is exactly a separable problem with two hard blocks: the session model and the action coverage. Generation 1 with no crossover (one agent, one attempt) tends to nail one block and stub the other. The GA move is to spawn three variant implementations, score each by {compiles, matches the reference behaviour, survives the critic}, and then *crossover*: graft the best session model from one variant onto the best action-coverage from another. By Proposition 2 this reaches a correct tool in generations logarithmic in the block count, where one agent grinding a single lineage does not.

The three-layer controller. Stacking the loops of Definition 1 gives a single orchestrator: a **GA outer** loop (variants, critic-fitness, synthesis-crossover, annealed rounds) over a **ferromagnetic middle** loop (hold m near criticality, aim with the directive field) over an **MFG inner** loop (each variant’s agents best-respond to the shared mean). One knob — m / population diversity — threads all three, and the whole craft is annealing it from hot-diverse to critical-aligned without quenching to $m = 1$, with the directive aiming the descent. The companion paper builds the middle layer; this paper builds the outer.

8 What is real, what needs building, what is out of scope

Real. The GRPO group, its group-relative advantage, the AnyMoE expert-gate trainer, and the island decomposition are shipped and tested ([1], [2]); the harness already runs populations of agents with adversarial critics as gates and a loop-until-dry completeness check. Read as this paper prescribes, that is a genetic algorithm with selection and a fitness function.

Needs building. The two operators the harness under-uses are *crossover* (a synthesis agent that decomposes candidates and grafts modules, rather than tournament-selecting a single winner) and an explicit *temperature schedule* (annealing coupling, mutation, and selection pressure round over round with m measured each generation). Both are orchestration logic over the existing workflow engine, not model or kernel work.

Out of scope. We do not claim a literal energy function for a fleet, nor that every task is separable (Remark 1); deceptive or strongly non-separable objectives fall back to mutation-and-diversity and lose the crossover speedup. We do not treat multi-objective Pareto fronts, novelty search, or open-ended evolution here; the target is the decomposable engineering objective with a scalar gate, which is the fleet’s daily work.

9 Conclusion

A fleet run as independent search wastes its members’ partial wins; run as pure consensus it only averages. Run as a genetic algorithm it does the thing neither can: recombine. We placed selection, mutation, and crossover on one distribution p_t , showed selection is replicator drift and that frequency-dependent selection is the mean-field coupling of the companion game (Proposition 1), and isolated crossover as the operator responsible for the speedup — provably superlinear on the

separable landscapes the fleet actually faces (Proposition 2), with deception as the named boundary (Remark 1). Premature convergence turned out to be the companion paper’s ordered phase, so the two papers share one control variable and one prescription: anneal m toward criticality, never quench, and aim with the directive. The framework is not speculative: GRPO’s rollout group is the population, its group-relative advantage is selection, AnyMoE experts are the building blocks, and the harness’s critics are already the fitness. What remains is to name and add the two missing operators — a crossover-by-synthesis step and an annealing schedule — and let the fleet that trains the model, and the model that runs the fleet, climb the same landscape together.

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